MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E1

BODY AND BOATTAIL BUSHING FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all departments and agencies of the Department of Defense.

SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings for the fabrication and packing of body and boattail bushing for M7207E1 Warhead.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-106	Tape, Pressure-Sensitive, Adhesive Marking, Paper
UU-T-450	Tissue, Facial

MILITARY

MIL-A-48078	Ammunition Standard Quality Assurance provisions, General Specification for

STANDARDS

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-1169 Packaging, packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9332486	Body
9333753	Boattail Bushing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawings 9332486 and 9333753 all associated drawings and with all requirements of applicable standards and specifications.

3.3 Adhesion and mar resistance of paint.-The paint on the warhead shall not exhibit flaking, peeling, loss of adhesion, marring or discoloration.

3.4 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated in a thorough, workmanlike manner. The parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, dents, tool marks, chipped edges, porosity, warpage, gouges, surface defects, dirt, grease, corrosion products, burn marks, and other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. Surface coatings shall be continuous except for scratches not exposing base metal.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted.

a. First article inspection.

b. Quality conformance inspection.

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article sample shall consist of the following item in sample quantities as indicated.

Part Description	Drawing	Quantity
Body	9332486	20
Boattail bushing	9333753	20

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Tow Warhead M207E1 Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9332486)				
	Examination for Defects	20	-	3.2	4.4.2.1
	Adhesion and Mar Resistance of Paint	20	-	3.3	4.5.2
	Boattail Bushing (Dwg. 9333753)				
	Examination for Defects	20	-	3.2	4.4.2.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Body (dwg. 9332486) from not more than one interfix lot number from one manufacturer.

b. Boattail bushing (dwg. 9333753) from not more than one interfix lot number from one manufacturer.

4.4.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63388

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Body				9332486
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Large inside diameter		0.40%	3.2	Gage
102.	True position of small holes		100%	3.2	Gage
103.	Diameter of small holes		0.40%	3.2	Gage
104.	Perpendicularity of aft shoulder with outside dia.		0.40%	3.2	Gage
105.	Large aft diameter		100%	3.2	Gage
106.	Runout of large aft diameter with datum		0.40%	3.2	Gage
107.	Pitch diameter of large aft thread		0.40%	3.2	Gage
108.	Length of ring retaining full thread		0.40%	3.2	Gage
109.	True position of small aft holes		100%	3.2	Gage
110.	Pitch diameter of small internal threads		0.40%	3.2	Gage
111.	Location of wireway slot		0.40%	3.2	Gage
112.	Location of boattail positioning slot		0.40%	3.2	Gage
113.	Wall thickness		0.40%	3.2	Gage
114.	Small inside diameter at aft end		0.40%	3.2	Gage
115.	Runout of small inside diameter at aft end with datums		0.40%	3.2	Gage
116.	Inside corner radius, large aft mounting shoulder		0.40%	3.2	Gage
117.	Adhesion and mar resistance of paint		0.10%	3.3	4.5.2

NOTES:

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QUALITY CONFORMANCE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9332486
4.4.2.1	Body				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Distance from front face to large dia. aft base		0.65%	3.2	Gage
202.	Distance from front face to shoulder		0.65%	3.2	Gage
203.	Distance from front face to inside shoulder		0.65%	3.2	Gage
204.	Large datum location aft end		0.65%	3.2	Gage
205.	Small datum location aft end		0.65%	3.2	Gage
206.	Minor diameter of small aft threads, max.		0.65%	3.2	Gage
207.	Major diameter of large aft threads, min.		0.65%	3.2	Gage
208.	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63388 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Boattail bushing				9333753
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Width of Warhead positioning tab		0.40%	3.2	Gage
102.	Height of Warhead positioning tab		0.40%	3.2	Gage
103.	Width of S&A positioning tab		0.40%	3.2	Gage
104.	Height of S&A positioning tab		0.40%	3.2	Gage
105.	Height of boattail bushing flange		0.40%	3.2	Gage
106.	Inside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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4.4.3 Testing. Testing is described in First Article and Quality Conformance inspection tables.

4.4.4 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 and 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of Inspection

4.5.1 Materials, Components and Processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Adhesion and mar resistance tests

4.5.2.1 Adhesion.-Firmly apply one (1) inch wide strip of masking tape conforming to UU-T-106 for the full length of one (1) of the launch leader bonding areas and peel the tape off in one abrupt motion.

4.5.2.2 Mar resistance.-Lay a small pad of tissue, conforming to the requirements of UU-T-450, approximately one (1) inch square and 8 to 12 layers thick, on one (1) of the launch leader bonding areas, other than the arch subjected to the adhesion test. Fully saturate the pad with thinner utilized in the painting process. Cover the pad with a small beaker. Remove the pad after one (1) minute and rub the moistened spot with a clean dry cloth or tissue. Failure is indicated by discoloration of the cloth or tissue or by visible evidence of damaged to the painted surface. A slight whitening of the surface which may be removed by light rubbing with a dry cloth shall not be reason for rejection.

5. PACKAGING

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5.1 Packaging and Packing. The parts shall be packaged and packed in accordance with the requirements of MIL-STD-1169.

5.2 Marking. Marking shall be in accordance with the requirements of MIL-STD-1169.

NOTES

6.1 Intended Use. The body and boattail bushing covered by this specification are components of the Improved TOW Warhead Section, Guided Missile, HE, M207E1.

6.2 Ordering Data. See MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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Preparing Activity
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